THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $2\,HCl + Ca(OH)_2 \longrightarrow CaCl_2 + 2\,H_2O$, the salt formed is: 1

(a) calcium carbonate **(b)** calcium sulphate
(c) calcium nitrate **(d)** calcium chloride

2. Why does a tooth paste contain a base? 1

(a) to colour teeth **(b)** to make teeth sour
(c) to neutralise mouth acids **(d)** to sweeten

3. A sample turns turmeric red and turns China rose green. It must be: 1

(a) basic **(b)** acidic **(c)** salt **(d)** neutral

4. Which of the following is a neutralisation reaction? 1

(a) $HCl + KOH \longrightarrow KCl + H_2O$ **(b)** $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ **(c)** $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ **(d)** $CaCO_3 \longrightarrow CaO + CO_2$

5. Why is dilute acid used in labs, not concentrated? 1

(a) concentrated acid is dangerous **(b)** dilute is expensive
(c) dilute is coloured **(d)** dilute is basic

6. After adding NaOH to HCl until the litmus remains violet, the solution is: 1

(a) basic **(b)** sour **(c)** neutral **(d)** acidic

7. Which of the following is not produced in neutralisation? 1

(a) hydrogen gas **(b)** heat **(c)** water **(d)** salt

8. Why is a few drops of acid added to water and not water into acid? 1

(a) the acid will evaporate **(b)** heat produced can splash the acid
(c) water is basic **(d)** acid is lighter than water

9. More NaOH is added to the neutral mixture above. Red litmus will turn: 1

(a) green **(b)** blue **(c)** yellow **(d)** red

10. Which of the following is the best way to treat a wasp sting (alkaline)? 1

(a) vinegar **(b)** soap **(c)** baking soda **(d)** lime water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name two ways in which a base is used to correct an acidic condition. 2

12. Explain why an acid bottle should be handled carefully. 2

13. Explain why a mixture of vinegar and baking soda bubbles. 2

14. Why does the colour of China rose change from pink to green when soap is added? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how neutralisation helps in curing acidity and in soil treatment. 3

16. Differentiate between a neutral and an acidic solution using indicators. 3

17. How would you identify an unknown colourless solution as acidic, basic or neutral without tasting? 3

OR

Write the balanced equations for the reaction of HCl with KOH and H_2SO_4 with KOH. 3

18. Explain how you will test a sample of rainwater and a sample of tap water with litmus. 3

OR

Why does lime water turn red litmus blue but vinegar does not? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how acids and bases are used in industries and at home, giving three examples each. 5

OR

Explain how a person can use household items to test the soil of a garden for acidity. 5

20. Describe how the colour changes of red cabbage juice show the nature of a solution and make a colour chart for common substances. 5

OR

Describe why factory effluents are neutralised before being released. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND